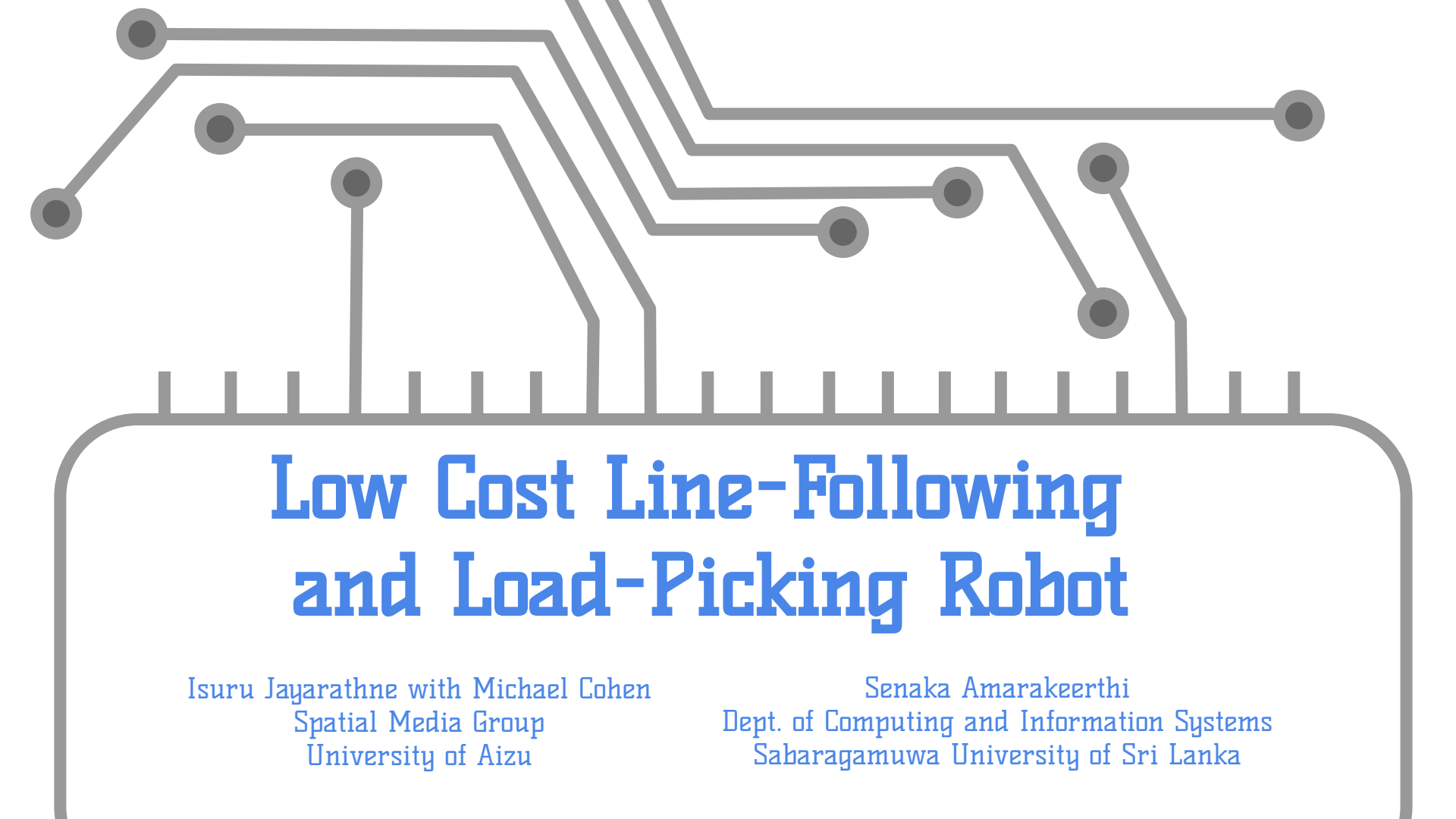




Low Cost Line-Following and Load-Picking Robot

Isuru Jayarathne with Michael Cohen
Spatial Media Group
University of Aizu

Senaka Amarakeerthi
Dept. of Computing and Information Systems
Sabaragamuwa University of Sri Lanka



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Background

- Robotics is an applied combination of mechanical engineering, electrical engineering, and computer science.
- Robots are controlled by microprocessors and use sensory feedback.
- Line-following is a navigation system for robots.
- Usually optical reflection is used to track colour difference between line and background and identify the path.

Past Research

- Line detection using infrared distance sensors to solve inclination problems (Ghani et al.)
- Line-following robot to detect line drawn on the floor (Pakdaman et al.)
- Line-following robot to deploy in shopping complexes for entertainment purposes (Colak et al.)
- Intelligent robot which can adjust the movement using magnetic sensors (Roman et al.)

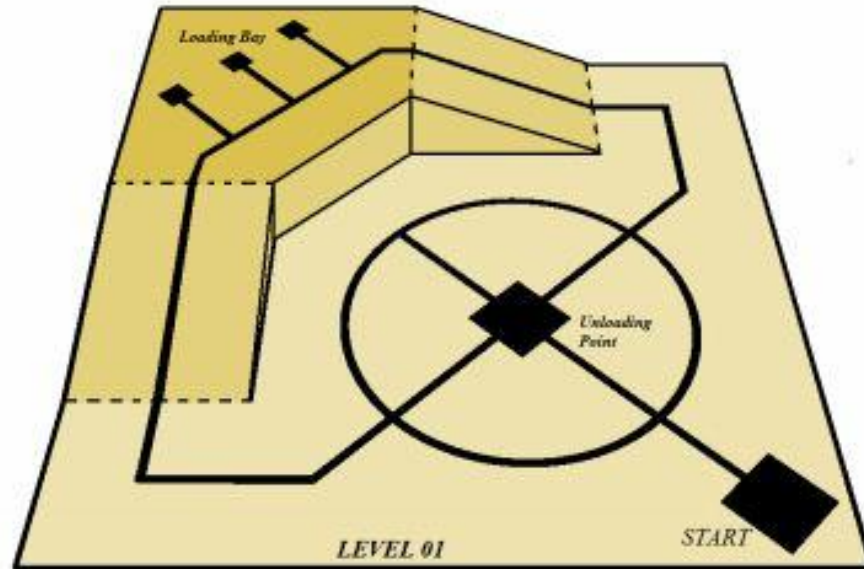


Motivation

- SAIM Robotics Challenge - 2014 (SRC '14) organized by South Asian Institute of Technology and Medicine (SAITM), Malambe, Sri Lanka.
- Automated line-following and load-picking robot

Motivation (Cont.)

- Arena of SRC '14 competition
- Consists of starting point, 3 loading points, and unloading



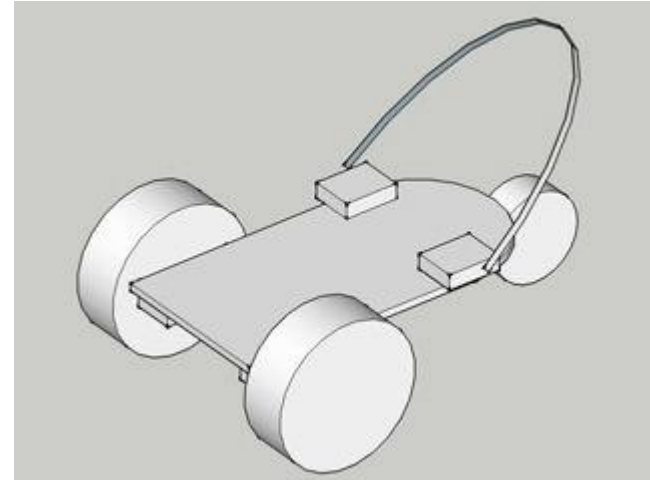


Goal

- Develop a line-following robot with inexpensive and minimal equipment for
 - Navigating along the lines and junctions
 - Finding the payload and placing it on unloading point
 - With power supply limited to 24 V

Approach (1/8)

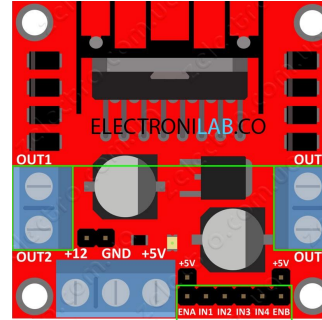
- Mechanical Parts
 - 2 gear motors
 - 1 omnidirectional wheel
 - 1 servomotor
 - Flexible plastic strip



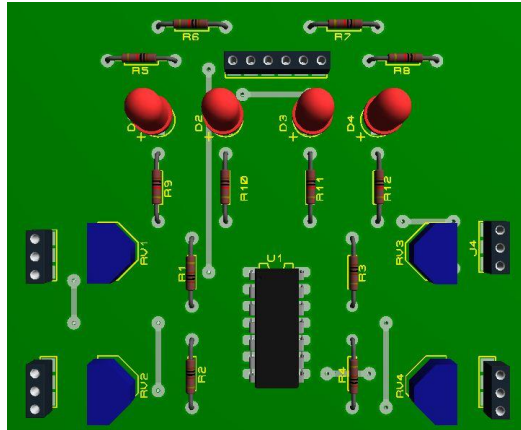
Approach (2/8)

- Circuitry
 - Arduino UNO processing board
 - HY-SRF 05 ultrasonic sensor module
 - 4 blue LEDs and 4 LDRs as line-detecting sensors
 - L298 dual full bridge motor control module
 - Comparator circuit module (LM324 integrated circuit, 4 presets, 4 LEDs, 4 10K resistors)

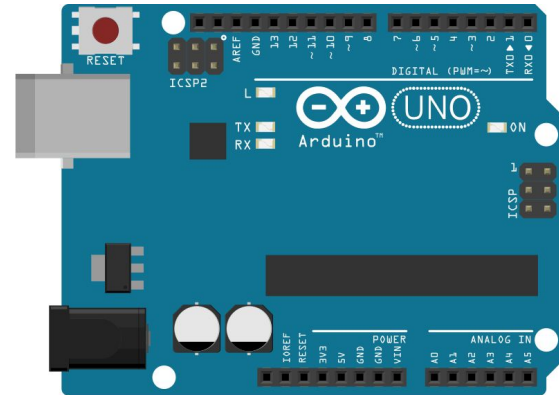
Approach (3/8)



L298 motor
control module



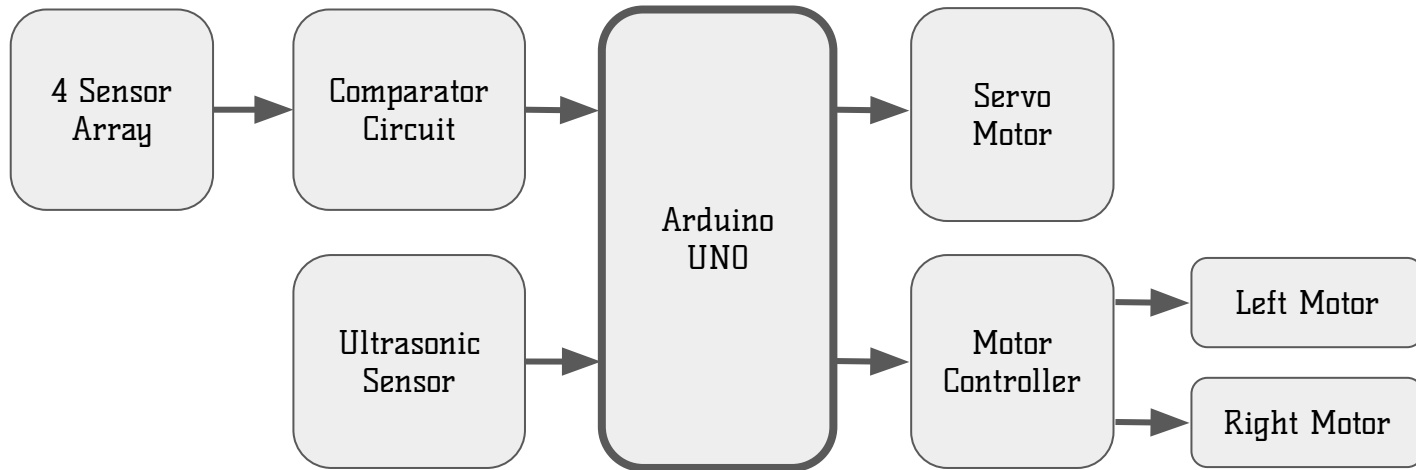
Comparator circuit



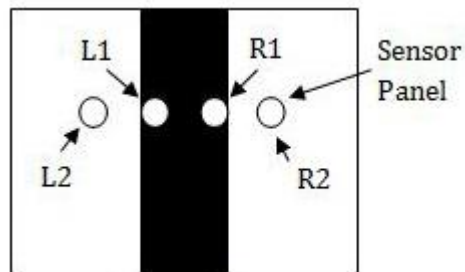
Arduino UNO

Approach (4/8)

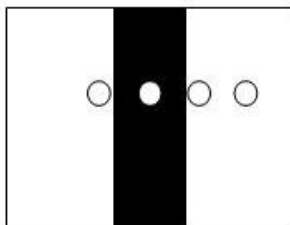
Block diagram of the project



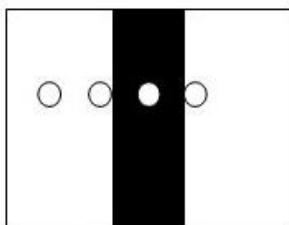
Approach (5/8)



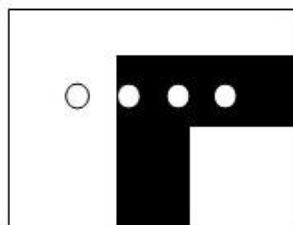
Move forward



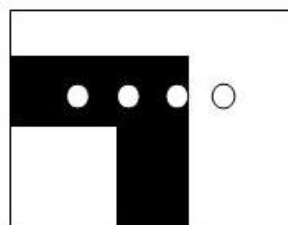
Turn left



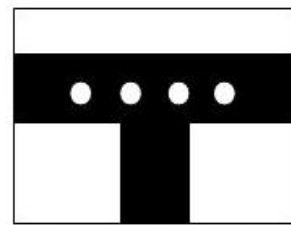
Turn right



Turn 90° right



Turn 90° left



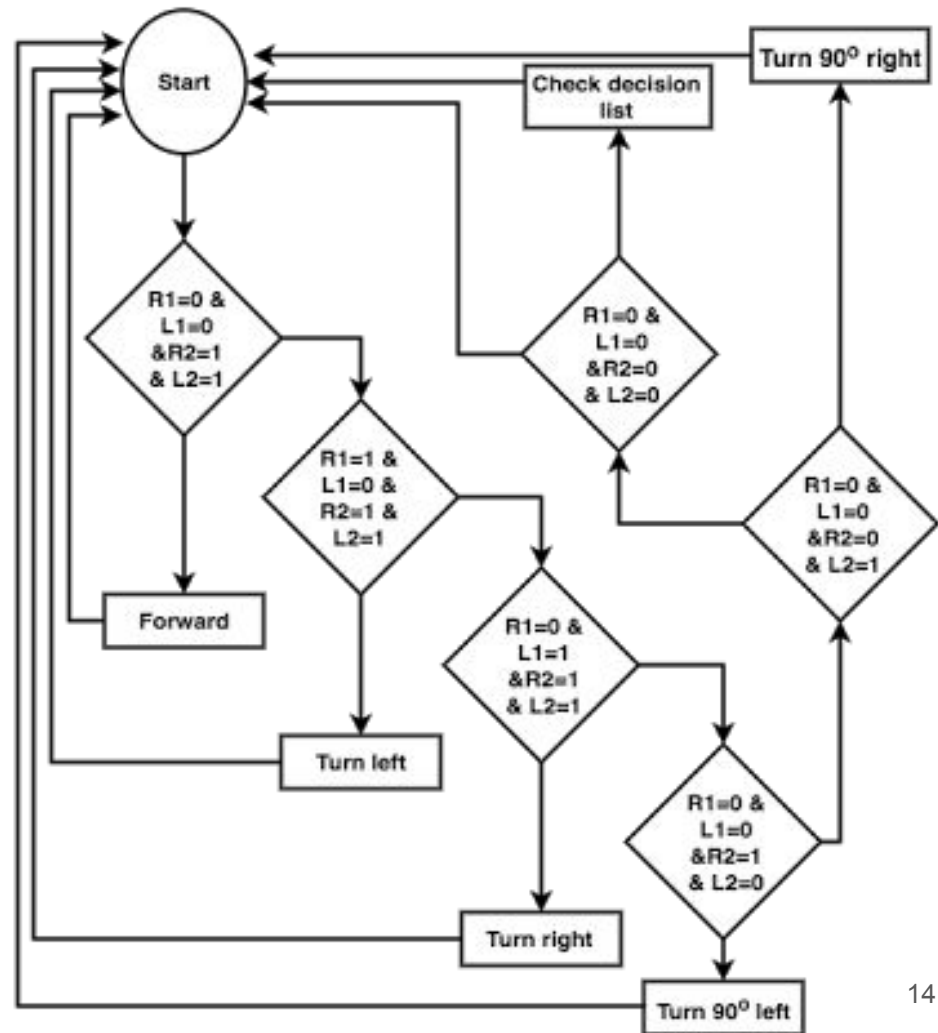
Determine operation
(from hard-coded
instruction list)

Approach (6/8)

- Line detection logic
 - When L1 and R1 both have digital value 0, robot moves forward.
 - When L1 has digital value 0, and others have digital value 1, robot turns left.
 - When R1 has digital value 0, and others have digital value 1, robot turns right.
 - When L2 has digital value 1, and others have digital value 0, robot turns 90° right.
 - When R2 has digital value 1, and others have digital value 0, robot turns 90° left.
 - When all sensors have digital value 0, the hard-coded decision list will be checked.

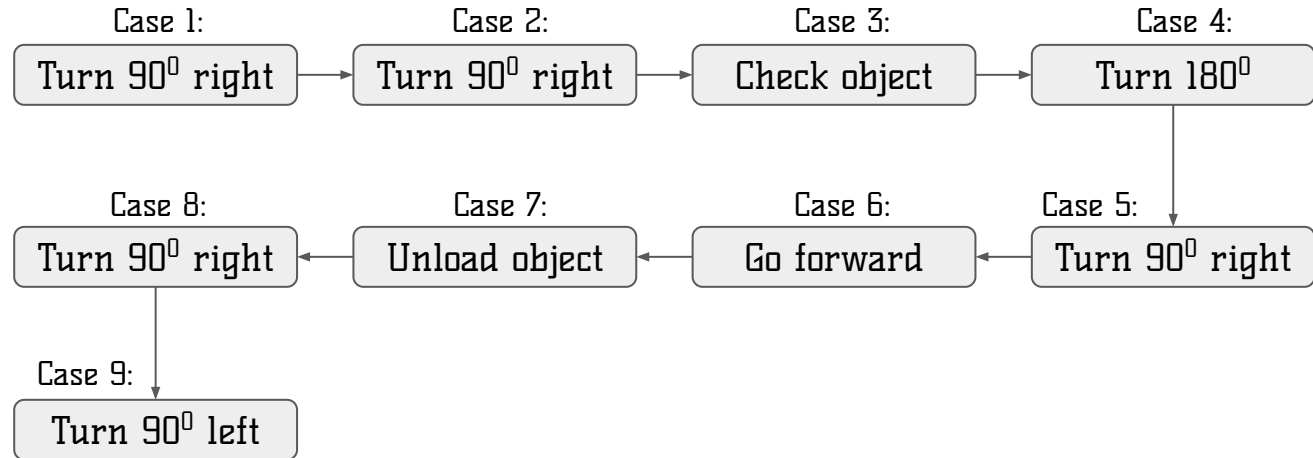
Approach (7/8)

Logic for follow
line with sensory
feedback

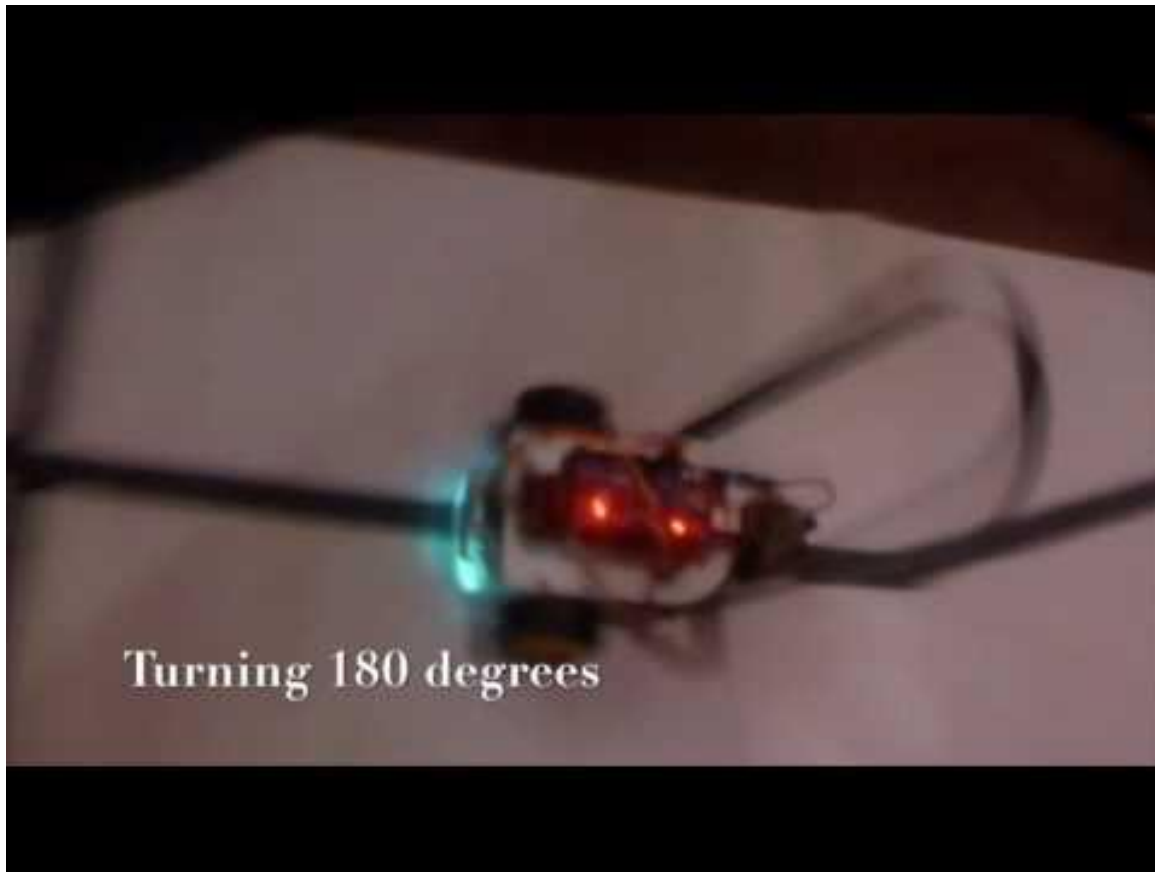


Approach (8/8)

- Order of the decision list was hard-coded into the Arduino board.



Video



Conclusion

- Successfully made turning decision according to hard-coded instructions.
- Robot could locate object located in random loading bay
- Robot has ability to make 90° turns, but shows poor performance in turning smoothly.
- Calibration is needed when light intensity changes.
- Payload-picking ability is satisfactory.
- Mechanical implementation should be fine-tuned to overcome these problems.



Future Works

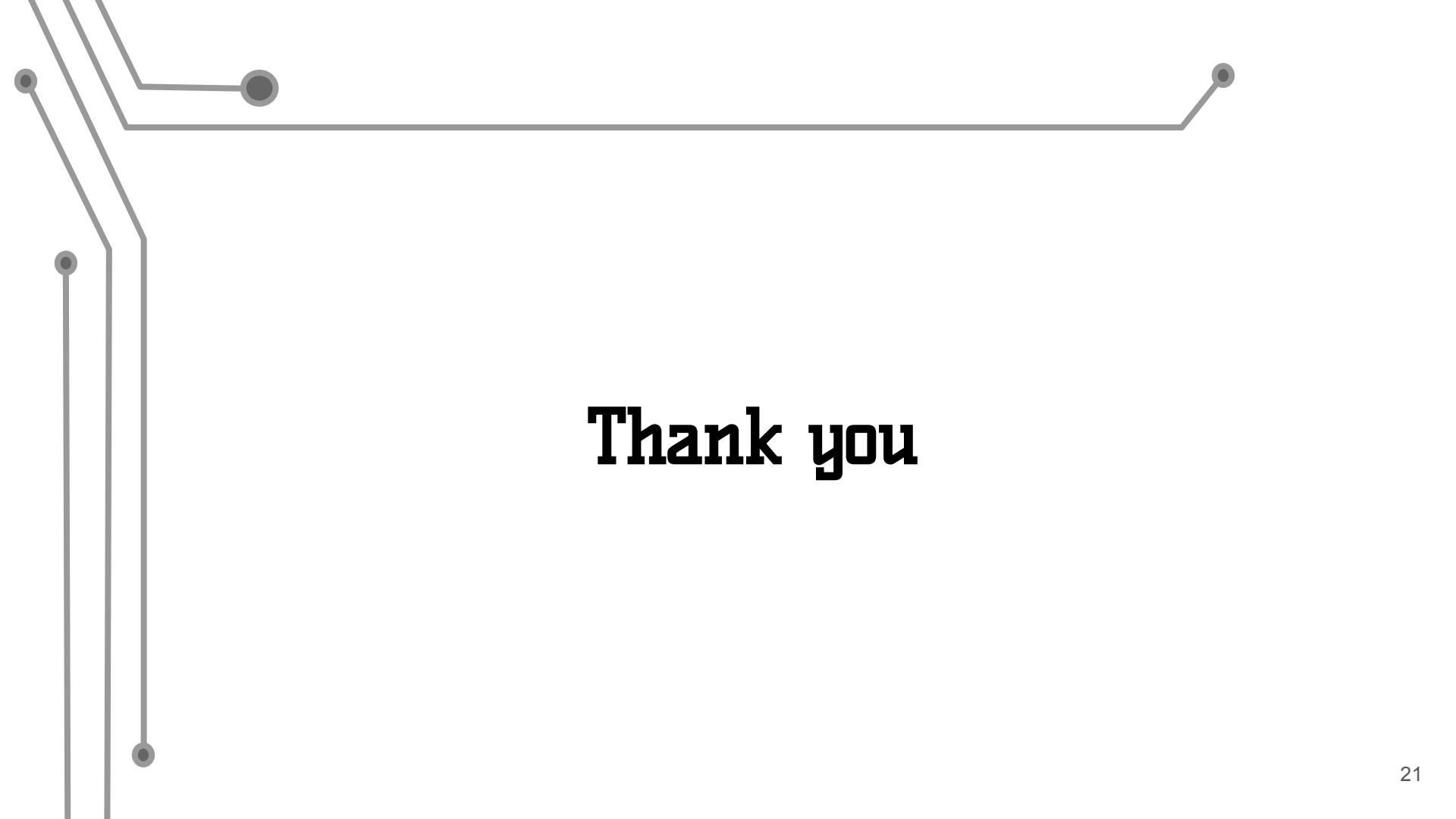
- Comparator circuit can be removed.
- L9110 H Bridge can use to decrease power consumption and weight.
- Autocalibration methods can be used.

References

- Tarneem Omar Barayyan and Enayyah Mohammed Barnawi. Enhanced line follower robot using IR coding as an artificial landmark. In *Int. Conf. in Recent Advances in Robotics and Mechatronics*, 2013.
- Ilknur Colak and Deniz Yildirim. Evolving a line following robot to use in shopping centers for entertainment. In *Industrial Electronics*, pages 3803–3807, 2009.
- Nor Maniha Abdul Ghani, Faradila Naim, and Tan Piow Yon. Two wheels balancing robot with line following capability. *World Academy of Science, Engineering and Technology*, pages 634–638, 2011.
- Takashi Gomi and Koichi Ide. Evolution of gaits of a legged robot. In *Fuzzy Systems Proceedings. IEEE World Congress on Computational Intelligence*, volume 1, pages 159–164, 1998.
- Peter Miller. Building a two wheeled balancing robot. *USQ Project*, 2008.

References (Cont.)

- Roman Osorio, Jose A Romero, Mario Pena, and Ismael Lopez-Juarez. Intelligent line follower mini-robot system. *Int. Journal of Computers, Communications & Control*, 1(2):73–83, 2006.
- Mehran Pakdaman and M Mehdi Sanaatiyan. Design and implementation of line follower robot. In *Computer and Electrical Engineering. ICCEE*, volume 2, pages 585–590. IEEE, 2009.
- Deepak Punetha, Neeraj Kumar, and Vartika Mehta. Development and applications of line following robot based health care management system. *Int. J. of Advanced Research in Computer Engineering & Technology (IJARCET)*, 2013.
- Juing-Huei Su, Chyi-Shyong Lee, Hsin-Hsiung Huang, Sheng-Hsiung Chuang, and Chih-Yuan Lin. An intelligent line-following robot project for introductory robot courses. *Science World Transactions on Engineering and Technology Education, Lunghwa University of Science and Technology, Taoyuan County, Taiwan*, 8(4):1–7, 2010.
- John-David Warren, Josh Adams, and Harald Molle. *Arduino for Robotics*. Springer, 2011.

A decorative graphic consisting of several gray lines that resemble circuit traces or a stylized network. The lines are primarily located on the left side and top of the slide, with some lines extending horizontally across the top. Small gray circles are placed at various points where the lines intersect or terminate, giving the impression of nodes or connection points in a circuit.

Thank you